

Lab 17 – AI for Data Processing: Data Cleaning and Preprocessing Scripts

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Task 1:

Prompt:

Develop a Python program to preprocess a loan dataset by handling missing values, encoding categorical variables, scaling numerical features, and splitting the dataset into training and testing sets. Use pandas for data handling and scikit-learn for scaling and train-test splitting. The program should prepare the dataset for machine learning model training.

Code & Output:

```
Assignment-17.5.py
Assignment-17.5.py >...
1 # Task- 1 :
2 import pandas as pd
3 from sklearn.preprocessing import StandardScaler
4 from sklearn.model_selection import train_test_split
5 # Sample dataset
6 data = {
7     'loan_amount': [10000, 15000, None, 20000, 25000],
8     'credit_score': [700, 750, 720, None, 680],
9     'loan_status': ['Approved', 'Rejected', 'Approved', 'Rejected', 'Approved']
10 }
11 df = pd.DataFrame(data)
12 print("\n Original Data:")
13 print(df)
14 # Handle missing values
15 df['loan_amount'] = df['loan_amount'].fillna(df['loan_amount'].mean())
16 df['credit_score'] = df['credit_score'].fillna(df['credit_score'].median())
17 print("\n After handling missing values:")
18 print(df)
19 # Encode loan_status
20 df['loan_status'] = df['loan_status'].map({'Approved': 1, 'Rejected': 0})
21 print("\n After encoding loan_status:")
22 print(df)
23 # Separate features and target
24 X = df[['loan_amount', 'credit_score']]
25 y = df['loan_status']
26 # Scale numerical features
27 scaler = StandardScaler()
28 X_scaled = scaler.fit_transform(X)
29 print("\n Scaled Features:")
30 print(X_scaled)
31 # Split dataset
32 X_train, X_test, y_train, y_test = train_test_split(
33     X_scaled, y, test_size=0.2, random_state=42)
34 )
35 print("\n Training Data:")
36 print(X_train)
37 print(y_train)
38 print("\n Testing Data:")
39 print(X_test)
40 print(y_test)
```

```

PS C:\Users\chara\OneDrive\Desktop\Ai-Assisted Coding> & C:
Original Data:
  loan_amount  credit_score  loan_status
0    10000.0         700.0    Approved
1    15000.0         750.0    Rejected
2      NaN         720.0    Approved
3    20000.0         NaN    Rejected
4    25000.0         680.0    Approved

After handling missing values:
  loan_amount  credit_score  loan_status
0    10000.0         700.0    Approved
1    15000.0         750.0    Rejected
2    17500.0         720.0    Approved
3    20000.0         710.0    Rejected
4    25000.0         680.0    Approved

After encoding loan_status:
  loan_amount  credit_score  loan_status
0    10000.0         700.0            1
1    15000.0         750.0            0
2    17500.0         720.0            1
3    20000.0         710.0            0
4    25000.0         680.0            1

Scaled Features:
[[-1.5      -0.51832106]
 [-0.5      1.64135001]
 [ 0.       0.34554737]
 [ 0.5     -0.08638684]
 [ 1.5     -1.38218948]]

Training Data:
[[ 1.5      -1.38218948]
 [ 0.       0.34554737]
 [-1.5     -0.51832106]
 [ 0.5     -0.08638684]]
4      1
2      1
0      1
3      0
Name: loan_status, dtype: int64

Testing Data:
[[-0.5      1.64135001]
 1      0]
Name: loan_status, dtype: int64
PS C:\Users\chara\OneDrive\Desktop\Ai-Assisted Coding>

```

Explanation:

This task demonstrates data preprocessing techniques required before training a machine learning model. Missing values are handled using mean and median methods, categorical data is encoded into numerical format, and numerical features are scaled using StandardScaler. Finally, the dataset is split into training and testing sets to prepare it for machine learning algorithms. This improves model accuracy and data quality.

Task 2:

Prompt

Create a Python program to clean a social media user dataset by removing duplicate records, handling missing values in numerical columns, and preparing the dataset for further analysis. Use pandas for data manipulation and ensure the dataset is cleaned and structured properly.:

Code & Output:

```
42 # Task- 2:
43 import pandas as pd
44 # Sample dataset
45 data = {
46     'user_id': [1, 2, 3, 4, 5, 1], # Duplicate user_id
47     'username': ['user1', 'user2', 'user3', 'user4', 'user5', 'user1'],
48     'age': [25, 30, None, 22, 28, 25],
49     'gender': ['Male', 'Female', 'Female', None, 'Male', 'Male'],
50     'platform': ['Instagram', 'Facebook', 'Twitter', 'Instagram', None, 'Instagram'],
51     'posts_count': [100, 150, 200, None, 250, 100],
52     'likes': [1000, 1500, None, 2000, 2500, 1000],
53     'engagement_level': ['High', 'Medium', 'Low', 'Medium', None,
54                          'High']
55 }
56 df = pd.DataFrame(data)
57 print("\n Original Data:")
58 print(df)
59 # Remove duplicate user profiles based on user_id
60 df = df.drop_duplicates(subset='user_id')
61 print("\n After removing duplicates:")
62 print(df)
63 # Handle missing values in numerical columns
64 df['age'] = df['age'].fillna(df['age'].mean())
65 df['posts_count'] = df['posts_count'].fillna(df['posts_count'].mean())
66 df['likes'] = df['likes'].fillna(df['likes'].mean())
67 print("\n After handling missing values in numerical columns:")
68 print(df)
```

```
PS C:\Users\chara\OneDrive\Desktop\Ai-Assisted Coding> & C:/Users/chara/AppData/Local/Programs/Python/Python311/Python.exe -i
Python 3.11.6 Shell
>
Original Data:
  user_id username  age  gender  platform  posts_count  likes  engagement_level
0        1   user1  25.0    Male  Instagram        100.0  1000.0             High
1        2   user2  30.0   Female   Facebook        150.0  1500.0             Medium
2        3   user3   NaN   Female   Twitter         200.0     NaN             Low
3        4   user4  22.0     NaN  Instagram         NaN  2000.0             Medium
4        5   user5  28.0    Male     NaN         250.0  2500.0             NaN
5        1   user1  25.0    Male  Instagram        100.0  1000.0             High

After removing duplicates:
  user_id username  age  gender  platform  posts_count  likes  engagement_level
0        1   user1  25.0    Male  Instagram        100.0  1000.0             High
1        2   user2  30.0   Female   Facebook        150.0  1500.0             Medium
2        3   user3   NaN   Female   Twitter         200.0     NaN             Low
3        4   user4  22.0     NaN  Instagram         NaN  2000.0             Medium
4        5   user5  28.0    Male     NaN         250.0  2500.0             NaN

After handling missing values in numerical columns:
  user_id username  age  gender  platform  posts_count  likes  engagement_level
0        1   user1  25.00    Male  Instagram        100.0  1000.0             High
1        2   user2  30.00   Female   Facebook        150.0  1500.0             Medium
2        3   user3  26.25   Female   Twitter         200.0  1750.0             Low
3        4   user4  22.00     NaN  Instagram        175.0  2000.0             Medium
4        5   user5  28.00    Male     NaN         250.0  2500.0             NaN
PS C:\Users\chara\OneDrive\Desktop\Ai-Assisted Coding>
```

Explanation:

This task focuses on data cleaning and preprocessing techniques such as removing duplicate records and handling missing values. Cleaning data improves data quality and ensures accurate analysis and machine learning results. Using pandas makes it easy to manipulate and preprocess large datasets efficiently. This step is essential in real-world data analysis and data science workflows

Task 3:

Prompt:

Develop a Python program to preprocess weather data by converting date columns to datetime format, handling missing values, performing feature engineering, and normalizing numerical values. Use pandas for data processing and MinMaxScaler for normalization.

Code & Output:

```
70 # Task-3:
71 import pandas as pd
72 from sklearn.preprocessing import MinMaxScaler
73 # Sample dataset
74 data = {
75     'date': ['2024-01-01', '2024-02-01', '2024-03-01', '2024-04-01', '2024-05-01'],
76     'temperature': [30, 28, None, 25, 22],
77     'humidity': [80, None, 75, 70, 65],
78     'rainfall': [5, 10, 15, None, 20]
79 }
80 df = pd.DataFrame(data)
81 print("\n Original Data:")
82 print(df)
83 # Convert date column to datetime format
84 df['date'] = pd.to_datetime(df['date'])
85 print("\n After converting date to datetime format:")
86 print(df)
87 # Handle missing values in temperature and humidity
88 df['temperature'] = df['temperature'].fillna(df['temperature'].mean())
89 df['humidity'] = df['humidity'].fillna(df['humidity'].mean())
90 print("\n After handling missing values in temperature and humidity:")
91 print(df)
92 # Feature engineering: Create season and week_number columns
93 df['season'] = df['date'].dt.month % 12 // 3 + 1 # 1: Winter, 2: Spring, 3: Summer, 4: Fall
94 df['week_number'] = df['date'].dt.isocalendar().week
95 print("\n After feature engineering:")
96 print(df)
97 # Normalize rainfall values using MinMaxScaler
98 scaler = MinMaxScaler()
99 df['rainfall_normalized'] = scaler.fit_transform(df[['rainfall']])
100 print("\n After normalizing rainfall values:")
101 print(df)
```

```

PS C:\Users\chara\OneDrive\Desktop\AI-Assisted Coding> & C:/Users/chara/AppData/Local/Programs/Python/Python311/Python.exe
Original Data:
  date temperature humidity rainfall
0 2024-01-01      30.0     80.0      5.0
1 2024-02-01      28.0     NaN     10.0
2 2024-03-01      NaN     75.0     15.0
3 2024-04-01      25.0     70.0     NaN
4 2024-05-01      22.0     65.0     20.0

After converting date to datetime format:
  date temperature humidity rainfall
0 2024-01-01      30.0     80.0      5.0
1 2024-02-01      28.0     NaN     10.0
2 2024-03-01      NaN     75.0     15.0
3 2024-04-01      25.0     70.0     NaN
4 2024-05-01      22.0     65.0     20.0

After handling missing values in temperature and humidity:
  date temperature humidity rainfall
0 2024-01-01      30.00     80.0      5.0
1 2024-02-01      28.00     72.5     10.0
2 2024-03-01      26.25     75.0     15.0
3 2024-04-01      25.00     70.0     NaN
4 2024-05-01      22.00     65.0     20.0

After feature engineering:
  date temperature humidity rainfall season week_number
0 2024-01-01      30.00     80.0      5.0      1         1
1 2024-02-01      28.00     72.5     10.0      1         5
2 2024-03-01      26.25     75.0     15.0      2         9
3 2024-04-01      25.00     70.0     NaN      2        14
4 2024-05-01      22.00     65.0     20.0      2        18

After normalizing rainfall values:
  date temperature humidity rainfall season week_number rainfall_normalized
0 2024-01-01      30.00     80.0      5.0      1         1      0.000000
1 2024-02-01      28.00     72.5     10.0      1         5      0.333333
2 2024-03-01      26.25     75.0     15.0      2         9      0.666667
3 2024-04-01      25.00     70.0     NaN      2        14      NaN
4 2024-05-01      22.00     65.0     20.0      2        18      1.000000

```

Explanation:

This task demonstrates feature engineering and data preprocessing techniques used in data science and machine learning. Converting date columns allows extraction of useful features like season and week number. Missing values are handled to maintain data consistency, and normalization ensures numerical values are scaled properly for machine learning models. These preprocessing steps improve model performance and data analysis accuracy.

Task 4:

Prompt:

Create a Python program to preprocess a movie dataset by removing duplicate records, handling missing values, encoding categorical variables, and normalizing rating values. Use pandas for data cleaning and MinMaxScaler for normalization.

Code & Output:

```
103 # Task-4 :
104 import pandas as pd
105 from sklearn.preprocessing import MinMaxScaler
106 # Sample dataset
107 data = {
108     'movie_id': [1, 2, 3, 4, 5, 1], # Duplicate movie_id
109     'title': ['Jersey', 'Eega', 'Pokiri', 'ok jaanu', 'Dhurandhar', 'HI nanna'],
110     'genre': ['Family', 'Drama', None, 'Rom-Com', 'Action', 'Family'],
111     'language': ['Telugu', 'Telugu', 'Telugu', None, 'Telugu', 'Telugu'],
112     'rating': [4.5, None, 3.0, 4.0, 5.0, 4.5]
113 }
114 df = pd.DataFrame(data)
115 print("\n Original Data:")
116 print(df)
117 # Remove duplicate movie entries based on movie_id
118 df = df.drop_duplicates(subset='movie_id')
119 print("\n After removing duplicates:")
120 print(df)
121 # Handle missing values in rating and genre
122 df['rating'] = df['rating'].fillna(df['rating'].mean())
123 df['genre'] = df['genre'].fillna(df['genre'].mode()[0])
124 print("\n After handling missing values in rating and genre:")
125 print(df)
126
127 # Encode categorical variables genre and language
128 df['genre_encoded'] = df['genre'].astype('category').cat.codes
129 df['language_encoded'] = df['language'].astype('category').cat.codes
130 print("\n After encoding genre and language:")
131 print(df)
132 # Scale rating values between 0 and 1 using MinMaxScaler
133 scaler = MinMaxScaler()
134 df['rating_normalized'] = scaler.fit_transform(df[['rating']])
135 print("\n After normalizing rating values:")
136 print(df)
```

```

PS C:\Users\chara\OneDrive\Desktop\Ai-Assisted Coding> & C:/Users/chara/AppData/Local/Programs/Python/Python313
Original Data:
movie_id  title    genre language rating
0 1 Jersey Family Telugu 4.5
1 2 Eega Drama Telugu NaN
2 3 Pokiri NaN Telugu 3.0
3 4 ok jaanu Rom-Com NaN 4.0
4 5 Dhurandhar Action Telugu 5.0
5 1 HI nanna Family Telugu 4.5

After removing duplicates:
movie_id  title    genre language rating
0 1 Jersey Family Telugu 4.5
1 2 Eega Drama Telugu NaN
2 3 Pokiri NaN Telugu 3.0
3 4 ok jaanu Rom-Com NaN 4.0
4 5 Dhurandhar Action Telugu 5.0

After handling missing values in rating and genre:
movie_id  title    genre language rating
0 1 Jersey Family Telugu 4.500
1 2 Eega Drama Telugu 4.125
2 3 Pokiri Action Telugu 3.000
3 4 ok jaanu Rom-Com NaN 4.000
4 5 Dhurandhar Action Telugu 5.000

After encoding genre and language:
movie_id  title    genre language rating genre_encoded language_encoded
0 1 Jersey Family Telugu 4.500 2 0
1 2 Eega Drama Telugu 4.125 1 0
2 3 Pokiri Action Telugu 3.000 0 0
3 4 ok jaanu Rom-Com NaN 4.000 3 -1
4 5 Dhurandhar Action Telugu 5.000 0 0

After normalizing rating values:
movie_id  title    genre language rating genre_encoded language_encoded rating_normalized
0 1 Jersey Family Telugu 4.500 2 0 0.7500
1 2 Eega Drama Telugu 4.125 1 0 0.5625
2 3 Pokiri Action Telugu 3.000 0 0 0.0000
3 4 ok jaanu Rom-Com NaN 4.000 3 -1 0.5000
4 5 Dhurandhar Action Telugu 5.000 0 0 1.0000

```

Explanation:

This task demonstrates data preprocessing techniques including duplicate removal, missing value handling, categorical encoding, and data normalization. Encoding categorical variables converts text data into numerical format for machine learning algorithms. Normalization scales rating values between 0 and 1 for better model performance. These preprocessing steps help prepare the dataset for machine learning and data analysis tasks.